

Aitezaz Sikandar Khan

Full - Stack Developer

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Professional Summary

Results-driven Software Engineer with a proven track record of shipping 10+ production-grade web applications. Specializing in high-performance React and Node.js architectures, I build scalable real-time systems and robust RESTful APIs. Adept at leveraging Docker, GitHub Actions, and AWS EC2 to accelerate deployments, delivering secure, optimized solutions that elevate user experiences.

Education

University of Peshawar

Bachelor of Science in Computer Science (GPA: 3.60 / 4.00)

Peshawar, Pakistan

2022 -- 2026

- **Achievements:** Secured the 95th percentile nationwide in the HEC National Skills Competency Test (NSCT) organized by HEC, PSEB, and P@SHA (April 2026).
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Networks, Software Engineering

Technical Skills

Languages: JavaScript, TypeScript, HTML5, CSS3, SQL

Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS, Material UI (MUI), Responsive Design

Backend: Node.js, Express.js, RESTful APIs, REST API Design, WebSockets (Socket.io)

Databases: MongoDB, Mongoose ODM, MySQL, Firebase

DevOps & CI/CD: Docker, GitHub Actions, AWS EC2, Nginx, Linux

Tools & Practices: Git, GitHub, Postman, Jest, Zod, Stripe API, Cloudinary, Agile/Scrum

Professional Experience

Freelance Full - Stack Developer

Independent Contractor

2025 -- Present

- Delivered 3 production MERN-stack applications end-to-end for local business clients (a complaint management portal, a personal finance ledger, and a content blog), handling requirements gathering, database design, deployment, and ongoing maintenance.
- Built a complaint tracking system that reduced manual follow-up by automating email notifications (Nodemailer), supporting media attachments (Cloudinary), and routing tickets via priority-based task queues.
- Implemented role-based access control with JWT authentication and Bcrypt hashing, protecting 15+ API endpoints and ensuring data isolation between user roles.

Projects

Collaborative Study Platform | React, Socket.io, Node.js, Gemini & Groq

- Developed a real-time virtual classroom utilizing Socket.io to synchronize slide changes, active users, and typing indicators with sub-100ms latency for up to 50 concurrent students.
- Designed a multi-LLM fallback mechanism that automatically redirects requests from Groq (Llama 3) to Google Gemini (3.5-Flash) when rate limits are hit, preventing AI chat interruptions.
- Eliminated audio latency by streaming text-to-speech voice narration directly to the browser using Express chunked transfer encoding headers.

Hospital Management System | React, Node.js, Express, MongoDB, Gemini API

- Built a secure, multi-role portal (Admin, Doctor, Patient) with appointment booking, using moment.js validation to prevent scheduling conflicts and Nodemailer for automated email reminders.
- Integrated an AI symptom-analysis assistant via the Gemini API, structuring prompt fallbacks to summarize patient history and generate relevant questions for doctor visits.
- Designed MongoDB aggregation pipelines to calculate monthly appointment distributions and doctor earnings, rendering the data on interactive dashboard charts.

DevOps-Enabled Blog Platform | React, Node.js, Docker, GitHub Actions, AWS EC2

- Packaged the React frontend and Node.js backend into multi-stage Docker containers, reducing image sizes via alpine bases to optimize deployment speed.
- Automated the deployment pipeline by writing a GitHub Actions workflow that builds Docker images and redeploys them to AWS EC2 via secure SSH commands.
- Configured Nginx as a reverse proxy to route public traffic, handle backend API requests, and serve compressed assets for reduced page load times.

E-Commerce Platform | Next.js, TypeScript, Redux, Stripe API, MongoDB

- Developed a fully typed storefront using Next.js App Router and TypeScript, managing persistent shopping cart states across pages with Redux Toolkit.
- Integrated Stripe Checkout for secure payments, implementing a backend webhook route to save orders to MongoDB and reset customer carts upon payment verification.
- Secured checkout forms and API endpoints against invalid data and malicious inputs using schema-based Zod validation and React Hook Form.